

OptArrow Requirements Specification

1. Introduction

The OptArrow optimization integration engine is designed to connect optimization clients with solver backends through a stable, high-performance transport layer. Current implementations utilize Python and Julia backends, with additional access provided through lightweight client interfaces such as MATLAB. The primary objective of OptArrow is to solve optimization problems, specifically Linear Programming (LP) and Quadratic Programming (QP). This document outlines the requirements guiding the intended architecture, technology choices, stakeholder engagement mechanisms, and future directions based on developer-confirmed evidence.

2. Stakeholder Engagement Mechanisms

2.1 Feedback Collection

- **Requirement:**
 - The system shall collect stakeholder feedback utilizing GitHub Issues as the primary mechanism for contribution and problem reporting.

2.2 Traceability

- **Requirement:**
 - The system shall implement a formal traceability mechanism to establish connections between stakeholder feedback, decisions, and implementation efforts in future releases.

3. Architectural Decision-Making Process

3.1 Technology Selection

- **Requirement:**
 - The architecture will utilize Apache Arrow for in-memory data transfer to support high performance and cross-language compatibility between Python and Julia.
 - The architecture will integrate HiGHS as a solver backend due to its open-source availability and popularity among scientific users.

4. Evolutionary Architecture Development

4.1 Iterative Development

- **Requirement:**
 - The architecture shall evolve iteratively, allowing flexibility for modifications based on stakeholder feedback and internal technological assessments.

4.2 Alternative Approaches

- **Requirement:**

- The architecture should revisit historical developments such as the COBRA Toolbox as reference points for evaluating current design choices.

5. Technology and Design Evaluation

5.1 Data Transfer Protocol

- **Requirement:**

- Alternative data-transfer protocols shall be evaluated through systematic trials to ensure optimal performance beyond the initial Apache Arrow implementation.

5.2 Design Alternatives

- **Requirement:**

- The development process shall explore multiple implementation routes to ensure robustness and adaptability to new challenges as they arise.

6. Process Governance and Maturity

6.1 Governance Structures

- **Requirement:**

- Future architecture governance will be structured around a formalized process, including comprehensive guidance on managing architectural changes.

6.2 Process Maturity

- **Requirement:**

- The project shall prioritize maturing processes such as requirements documentation, traceability, and decision analysis to drive consistency and reliability in future development cycles.

7. Future Directions and Recommendations

7.1 Stakeholder Feedback Enhancement

- **Proposed Requirement:**

- A formal process for reviewing stakeholder feedback and deriving actionable insights shall be developed, using industry-standard practices like Architecture Decision Records (ADRs).

7.2 Enhanced Traceability

- **Proposed Requirement:**

- The system will implement a traceability process that aligns stakeholder needs with decision and implementation documentation, enabling comprehensive requirements management.

Summary

This requirements specification ultimately lays the foundation for future developments in the OptArrow project by identifying essential components, processes, and technological considerations. The anticipated outcome is a more mature and robust optimization engine with enhanced scalability, process clarity, and stakeholder integration. Future iterations should aim to incorporate formal governance structures, traceability processes, and systematic evaluations of technological alternatives to continuously align with best practices in optimization software development.